RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

TITLE :SEAT SYNC YOUR HEALTH LINK

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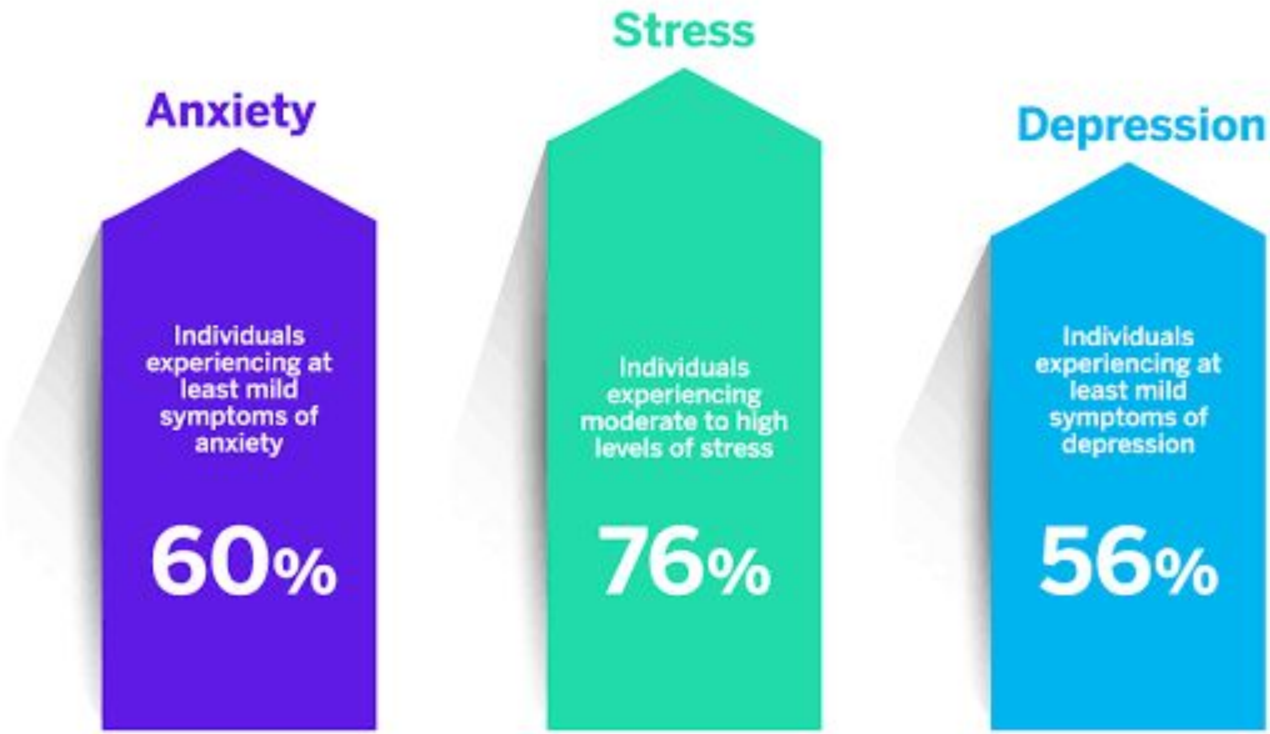
OUTLINE

- Abstract
- Objective
- Literature Survey
- Summary of Literature
- Proposed System & Novelty
- Block Diagram
- Hardware/Software requirements
- References

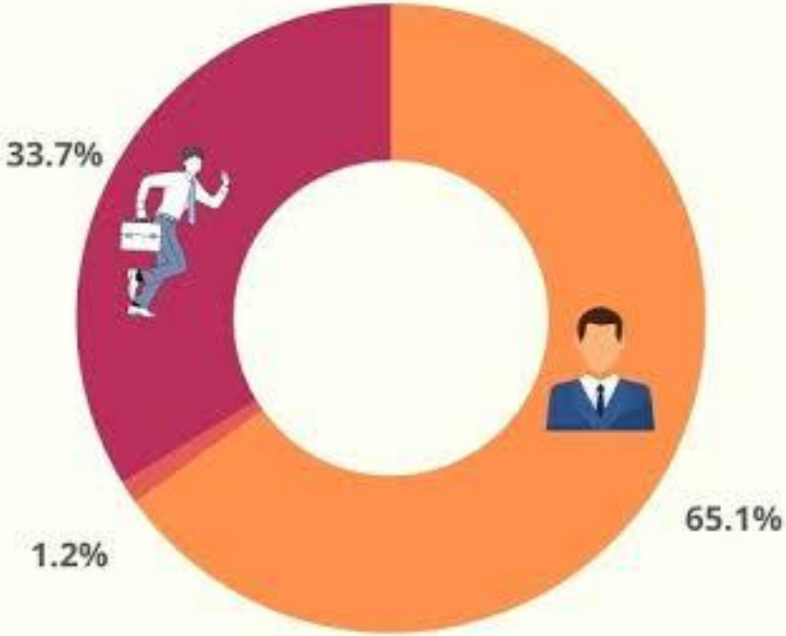
Mental Health & Wellbeing in 2023

qualtrics XM



Data from Champion Health. Sample size: 4170 individuals.

Should employers be concerned about their employee's mental health?



YES!
It has an impact on their productivity.

NO!
It's not their business. The employee has the duty to take care of their own mental health.

ALL HANDS ON DECK!
All parties need to play their part to create a mentally safe working environment.

Total number of respondents: 86

www.omtsdigest.com

Abstract

Technology Integration: Combines advanced sensor technology with furniture to enhance user well-being and productivity.

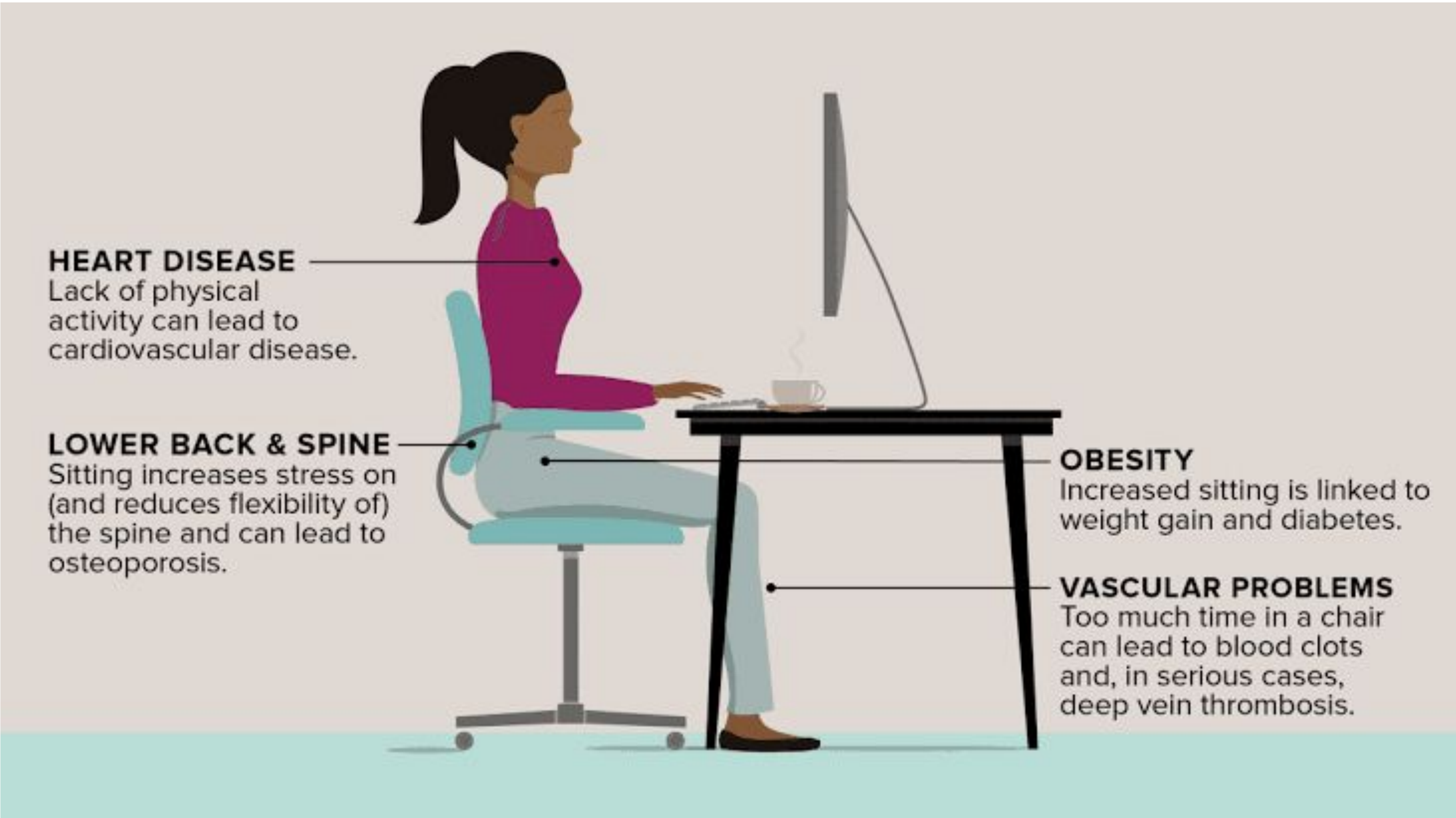
Sensor Suite: Utilizes heart rate monitors, occupancy detectors, load sensors, and temperature gauges to track sitting behaviors and physiological metrics.

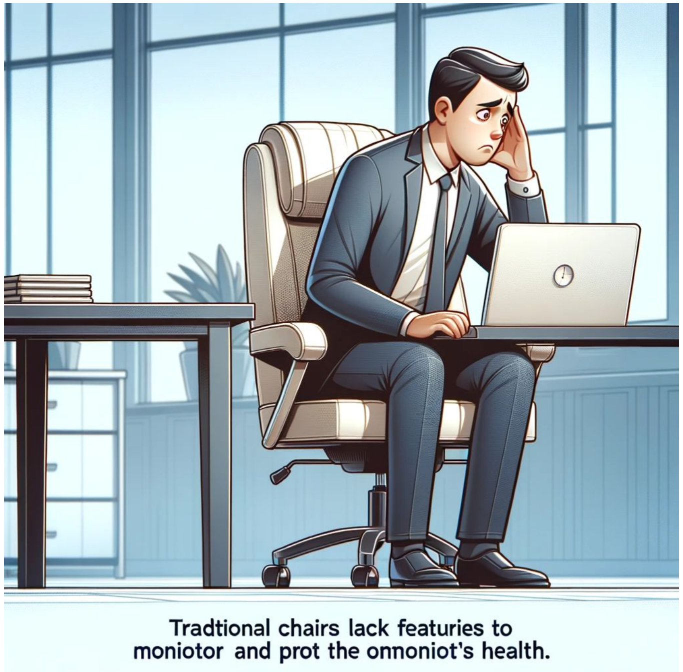
Data Analysis: Generates personalized feedback and alerts based on data analysis to prompt corrective actions.

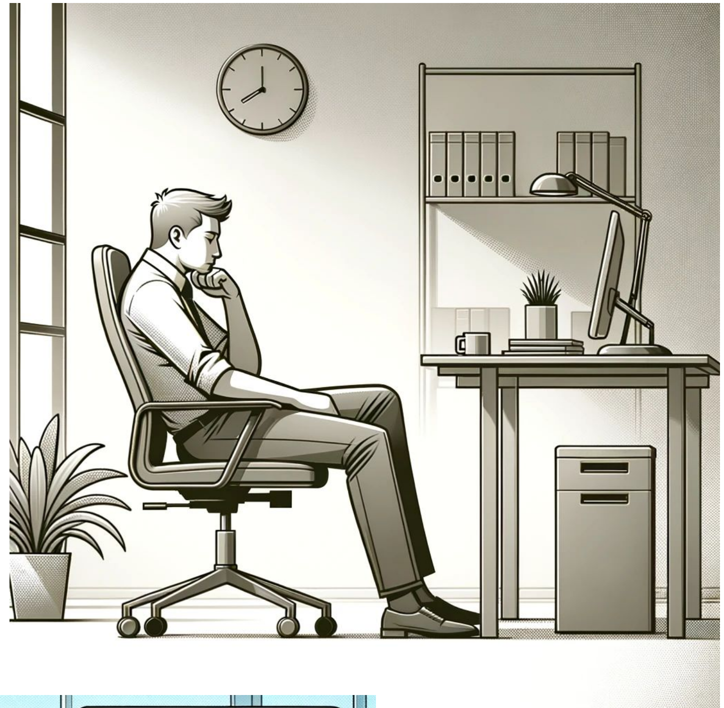
Gamification: Employs gamification techniques to incentivize positive behavior by rewarding good posture, breaks, and recommended sitting durations.

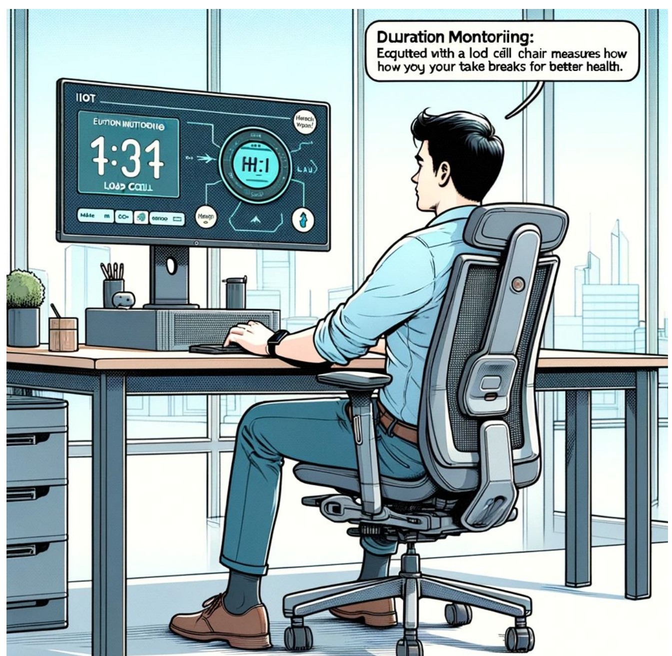
User Interface: Provides real-time feedback and progress tracking through a user-friendly dashboard accessible via web or mobile app.

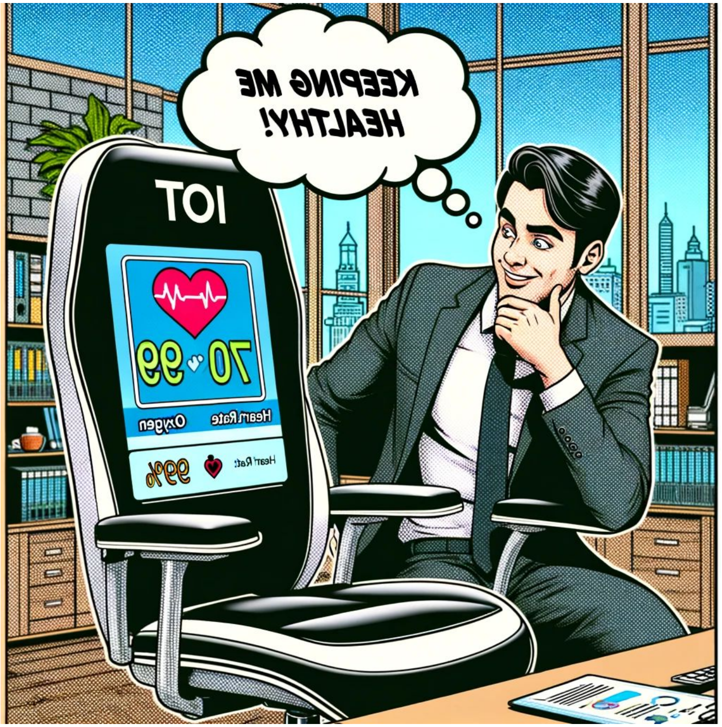
Health and Productivity: Aims to mitigate health risks associated with prolonged sitting and foster a dynamic, health-conscious workspace culture.


HEART DISEASE
Lack of physical activity can lead to cardiovascular disease.
LOWER BACK & SPINE
Sitting increases stress on (and reduces flexibility of) the spine and can lead to osteoporosis.
OBESITY
Increased sitting is linked to weight gain and diabetes.
VASCULAR PROBLEMS
Too much time in a chair can lead to blood clots and, in serious cases, deep vein thrombosis.

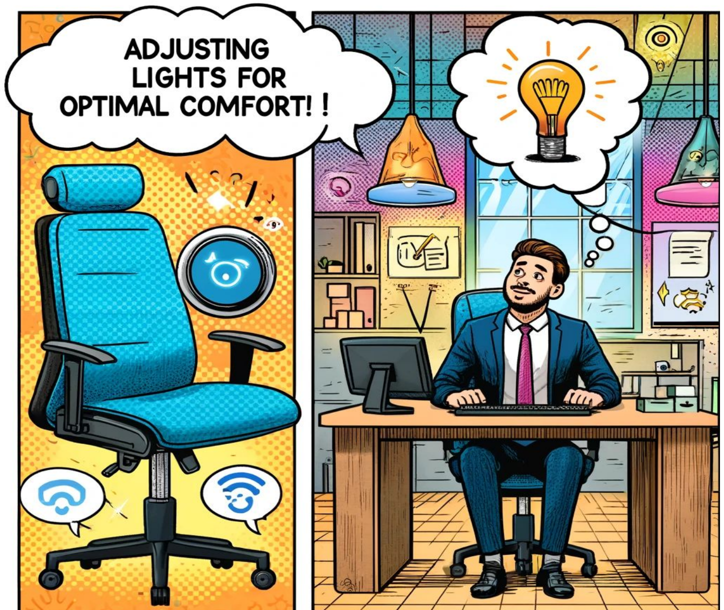

Tradtional chairs lack features to
monitor and prot the onnoniot's health.




Duration Montoriing:
Equtted with a lod cell chair measures how
how yoy your take breaks for better health.


KEEPING ME HEALTHY!


TIME FOR A QUICK SHRETCH!
SINUG


ADJUSTING LIGHTS FOR OPTIMAL COMFORT! !


AH, PERFECT SUPPORT!

Objective

Design and develop an IoT-enabled chair that uses sensor technology and gamification to promote healthier sitting habits and enhance user well-being in modern workspaces.

Key Goals:

- **Sensor Integration:**
  Use heart pulse, load, and ultrasonic sensors to monitor sitting habits and physiological state.

- **Data Analysis:**
  Identify patterns in behavior, heart rate, and temperature.

- **Alert System:**
  Trigger alerts for posture corrections and reduced sitting time.

- **Gamification:**
  Reward users for healthy sitting habits to encourage engagement.

- **Dashboard:**
  Provide real-time feedback, progress tracking, and recommendations via mobile app.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
IOT Based Smart Chair to help with Back pain	2023	Shilpa Shesham	Smart chair equipped with tactile pressure sensor mats and vibration actuators Data processing using simple rules and storage in a MongoDB database
IOT System for real time posture asymmetry detection	2023	Monica La Mura	Low-cost IoT measurement system with force sensing resistors (FSR) Java-based software with uncertainty-driven asymmetry detection algorithm
IoT enabled smart wheelchair solution for physically challenged people	2022	Biswajit Dwivedy	Smart wheelchairs with sensors and assistive technologies for mobility. Use of 8051 micro-controller and IoT technology with GPS and WiFi modules.

Summary of Literature

Posture Correction Chairs: Smart IoT chairs with sensors and wireless communication detect poor posture and notify users via visual and vibration feedback through a mobile app.

Continuous Monitoring: IoT chairs with backrest inclination sensors, humidity, and temperature modules continuously detect and notify users of incorrect sitting postures.

Smart Wheelchairs: IoT-based wheelchairs monitor health parameters, provide emergency notifications, and assist with sitting and standing using hydraulics.

Quadriplegic Support: IoT-enabled wheelchairs for quadriplegic patients feature hand gesture controls, alert systems, health regulation, and AI-based path prediction systems.

Proposed System

Key Components of the Proposed System

Sensor Integration:

- Heart pulse, load, and ultrasonic sensor for real-time data on user behavior and physiological state.

Data Analysis and Alerts:

- Analyze data for patterns and deviations; trigger alerts for corrective actions.

Gamification:

- Reward users with points and badges for healthy sitting habits to encourage engagement.

Health Dashboard:

- Web and mobile app providing real-time feedback, progress tracking, and personalized recommendations.

THRESHOLDS FOR MEASUREMENT

Component	Parameter	Normal	Warning	Critical
Load Cell with HX711	Sitting Duration	0 kg to 50 kg	51 kg to 100 kg	>100 kg
MAX30100 Sensor	Heart Rate	60 bpm to 100 bpm	101 bpm to 120 bpm	>120 bpm
MAX30100 Sensor	SpO2 Levels	95% to 100%	90% to 94%	<90%
BH1750 Sensor	Ambient Light	300 lx to 500 lx	100 lx to 299 lx or 501 lx to 800 lx	<100 lx or >800 lx

Novelty in Proposed System

Innovative Features of the Proposed System

Comprehensive Sensor Integration:
- Heart pulse, IR, load, and temperature sensors for real-time data on users' physiological state and sitting behaviors.

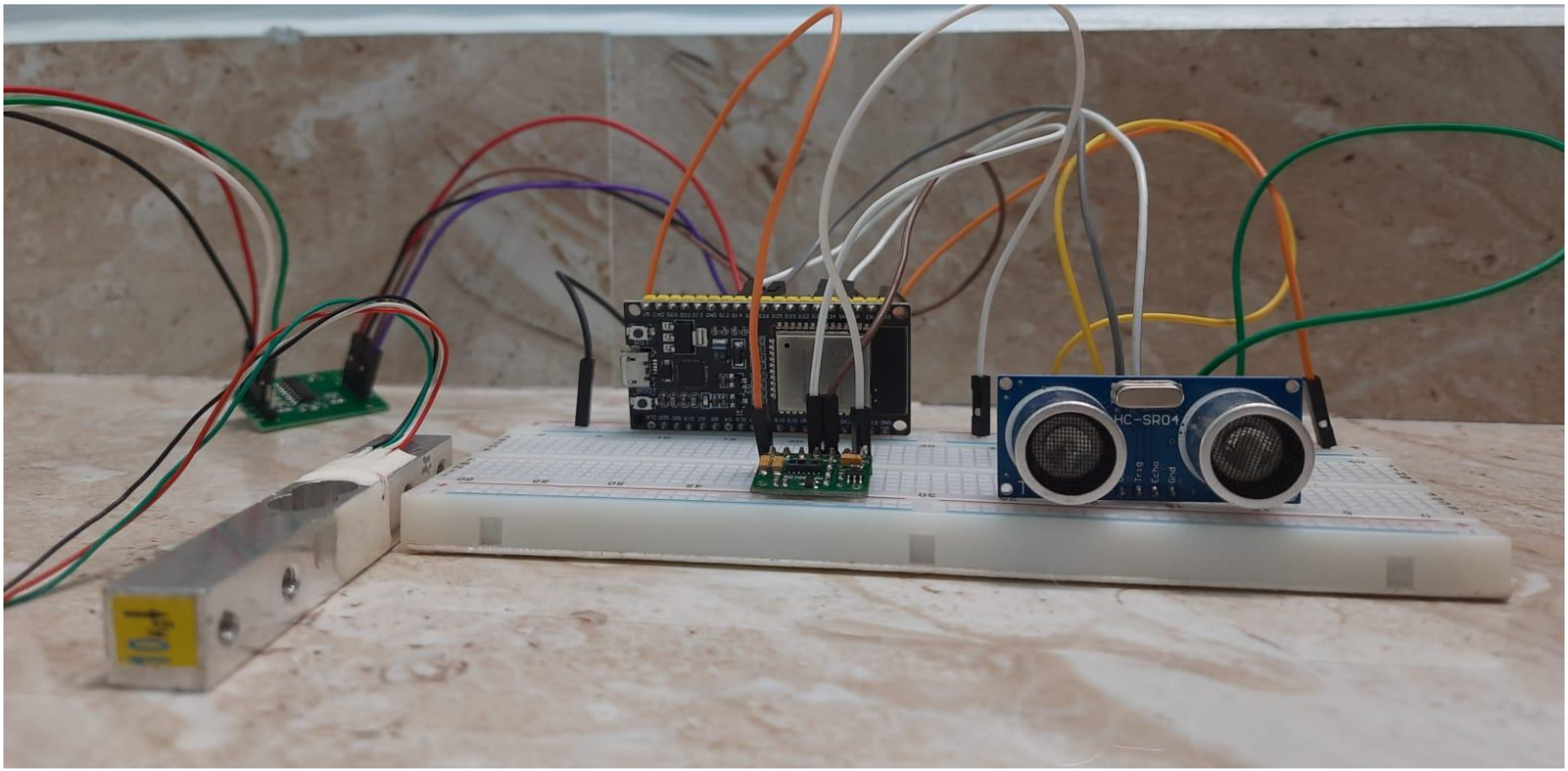
Advanced Data Analysis and Alerts:
- Prompt notifications for deviations from predefined thresholds to facilitate immediate corrective actions.

Gamification:
- Incentives such as points, badges, and rewards for maintaining healthy sitting habits, encouraging sustained engagement and behavior change.

User-Friendly Health Dashboard:
- Intuitive visualization of key metrics and achievements, accessible via web or mobile app, to enhance motivation and well-being.

This combination of sensor technology, gamification, and user-centric design offers a novel and effective approach to promoting healthier sitting habits and fostering a dynamic, health-conscious workspace environment.

Block Diagram

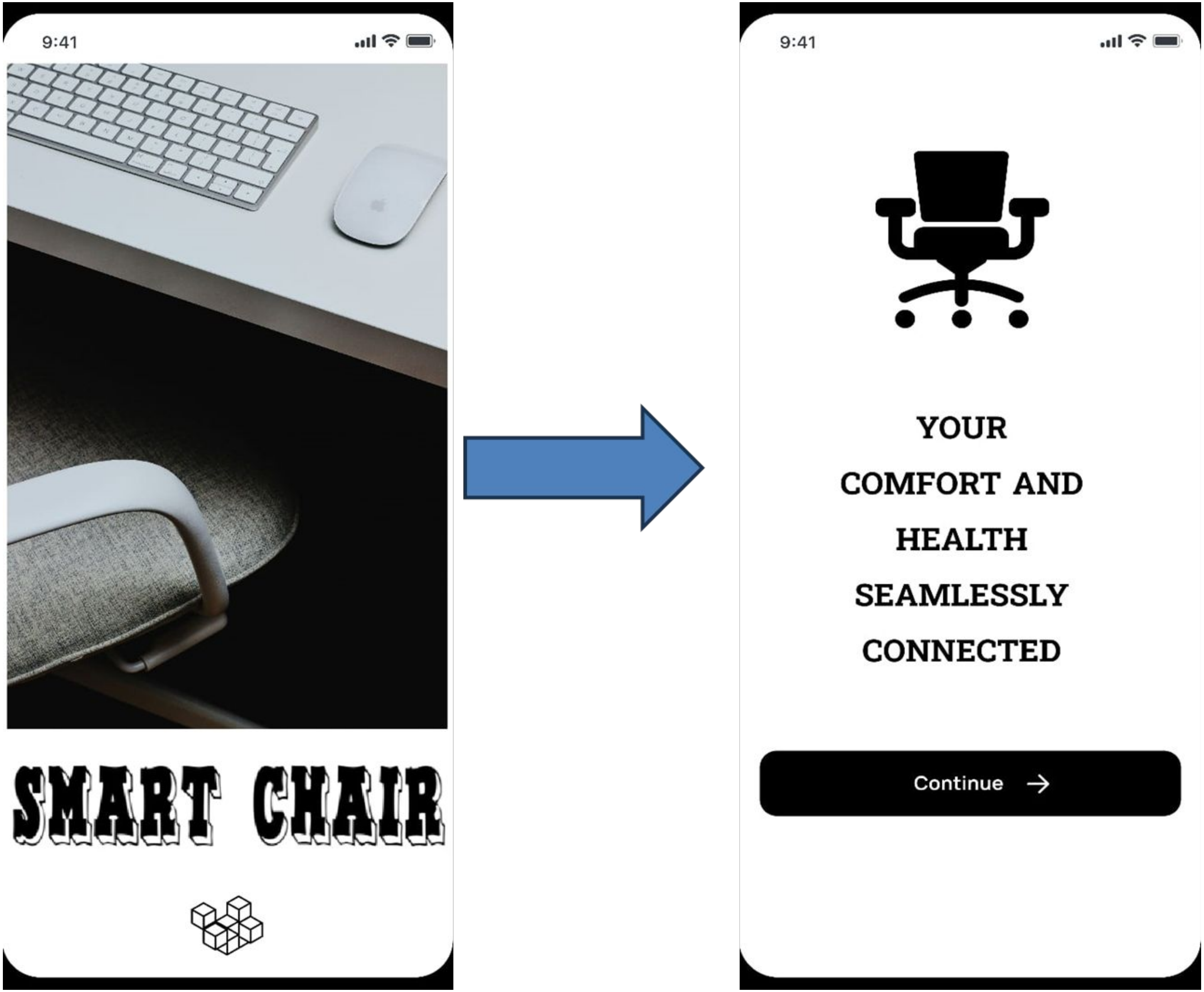

HC-SR04
Trig
Echo
Gnd

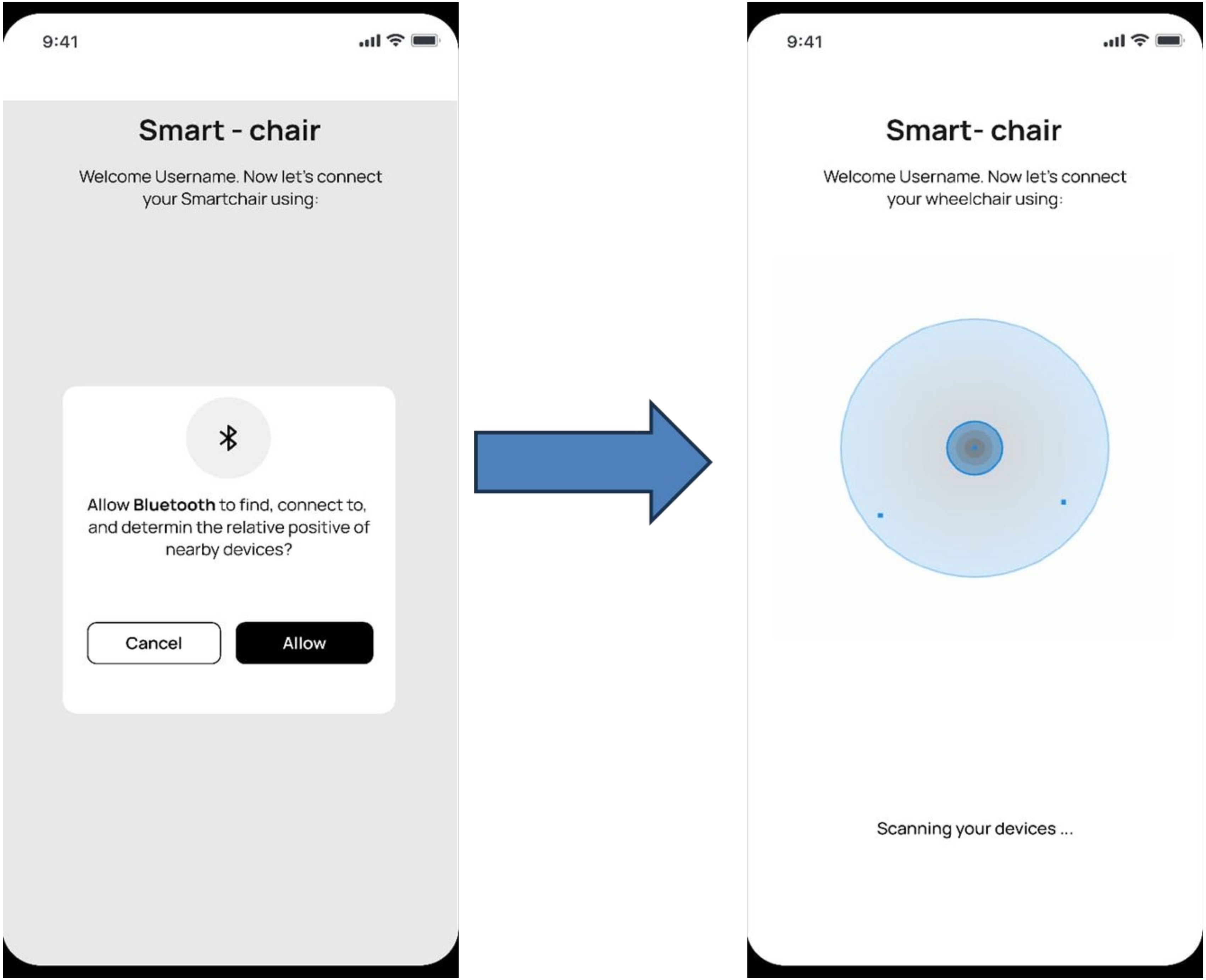
Hardware Requirements

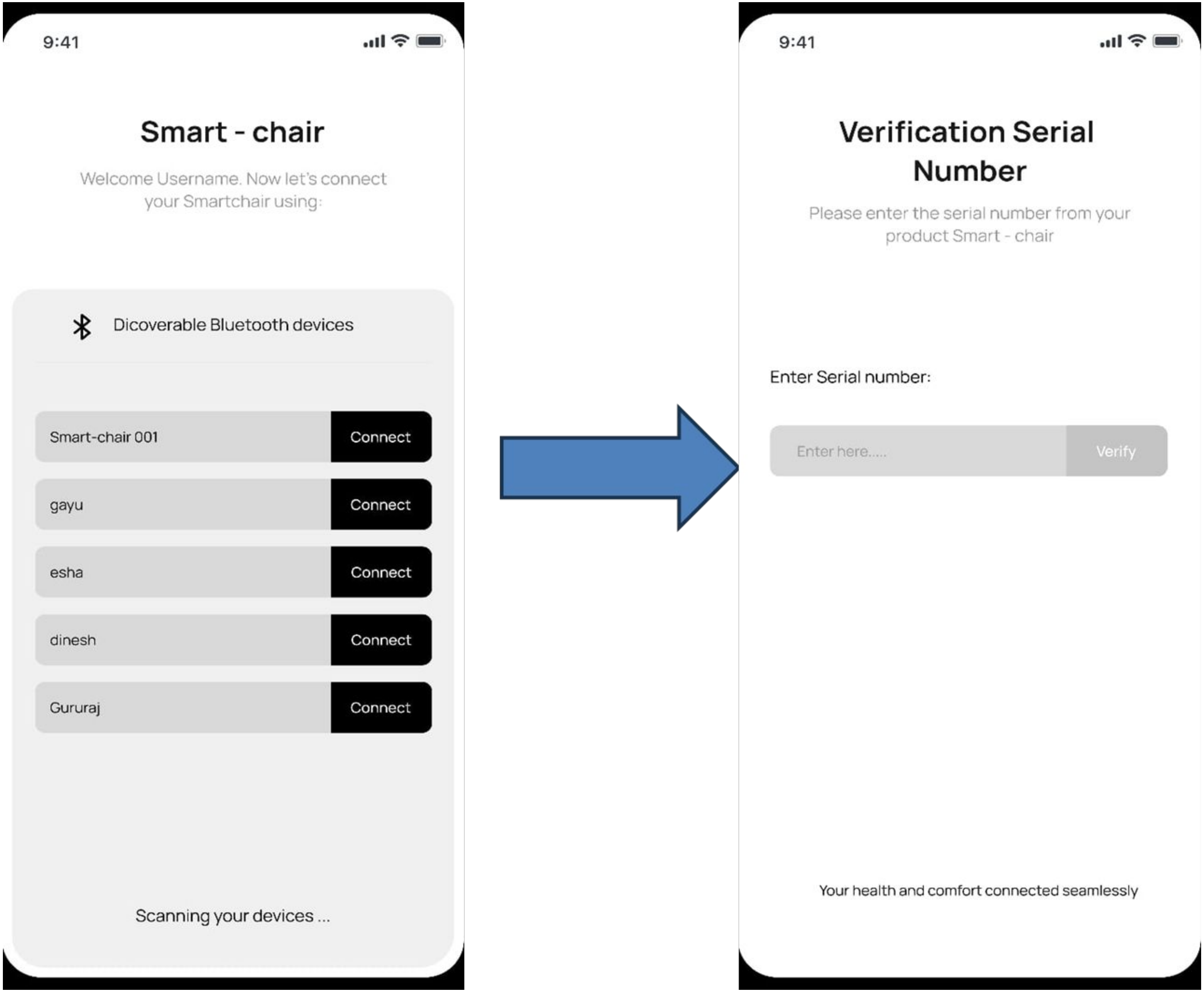
1.ESP32
2.Load Cell(HX711)
3.Ultrasonic Sensor
4. Heart Rate Sensor(MAX 30100)
5. Light ambient sensor(BH1750)

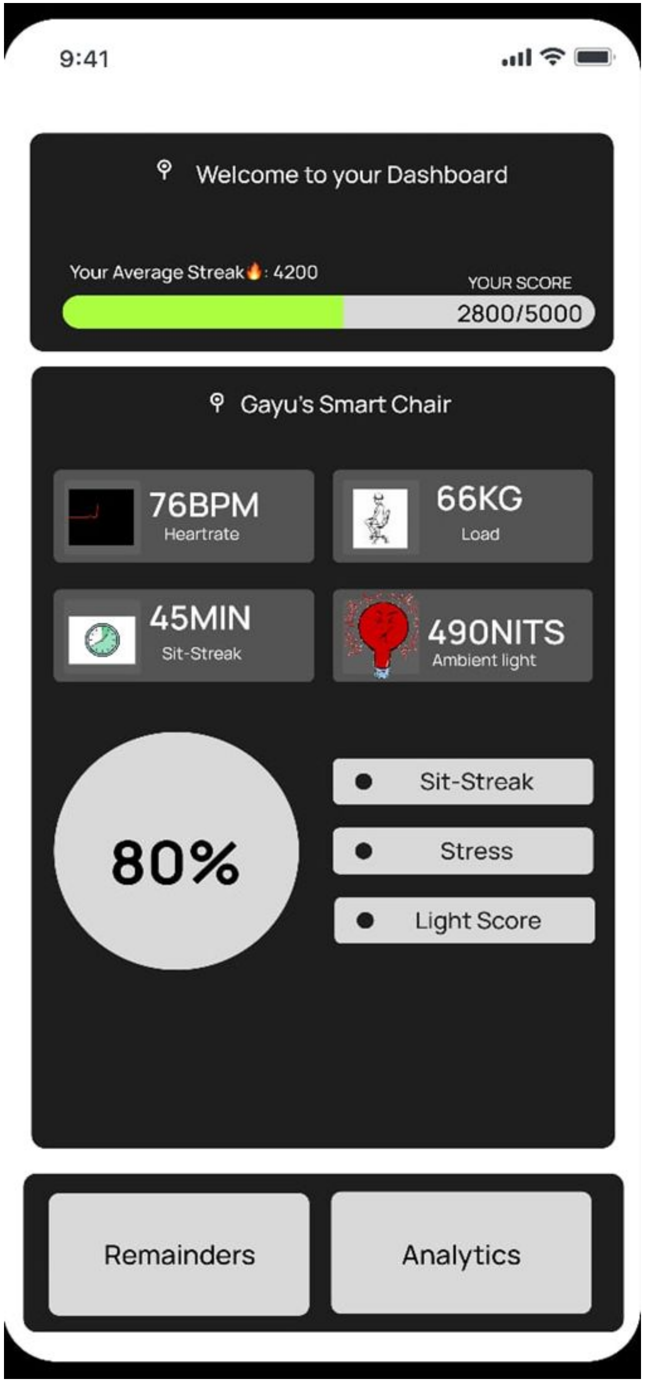
SOFTWARE REQUIREMENTS

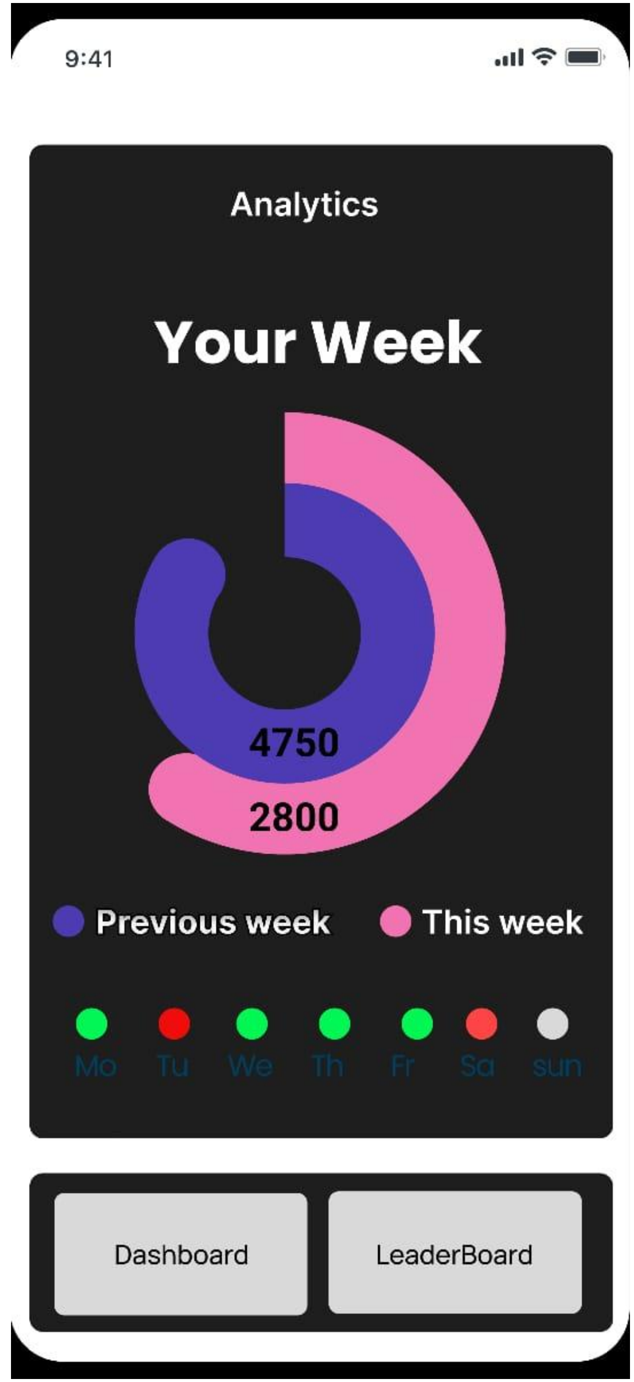
- Arduino IDE
- Blynk
- FIGMA

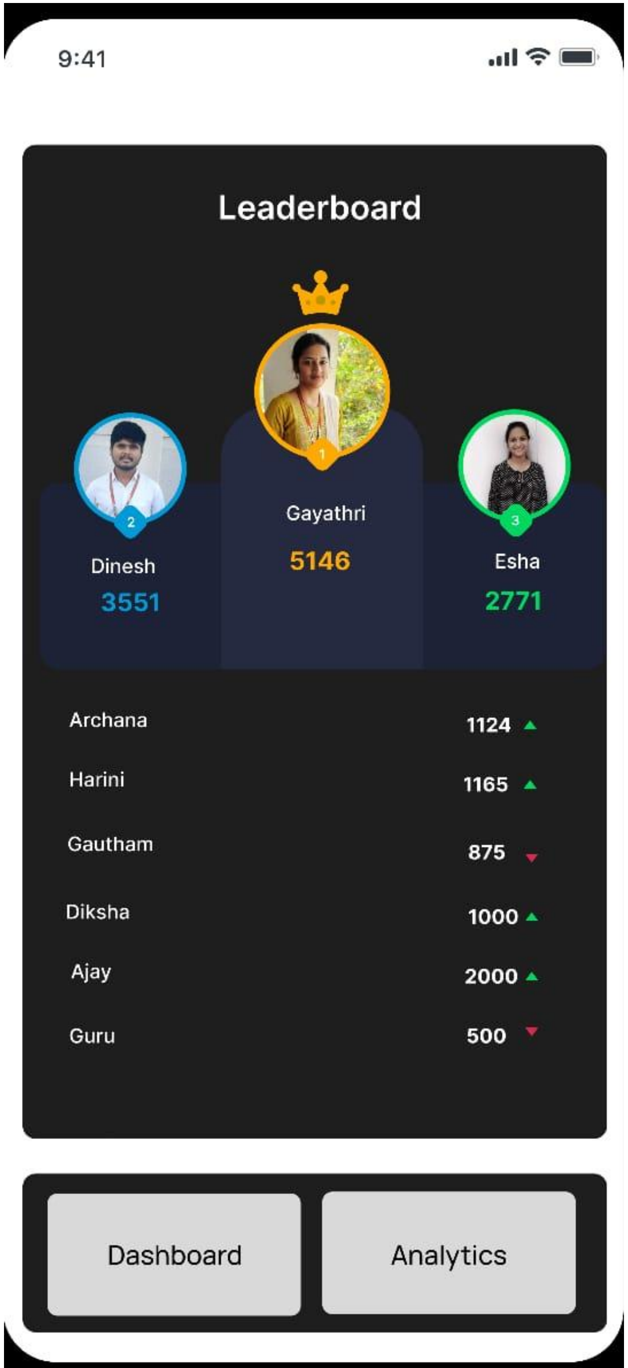

9:41
SMART CHAIR
9:41
YOUR
COMFORT AND
HEALTH
SEAMLESSLY
CONNECTED
Continue

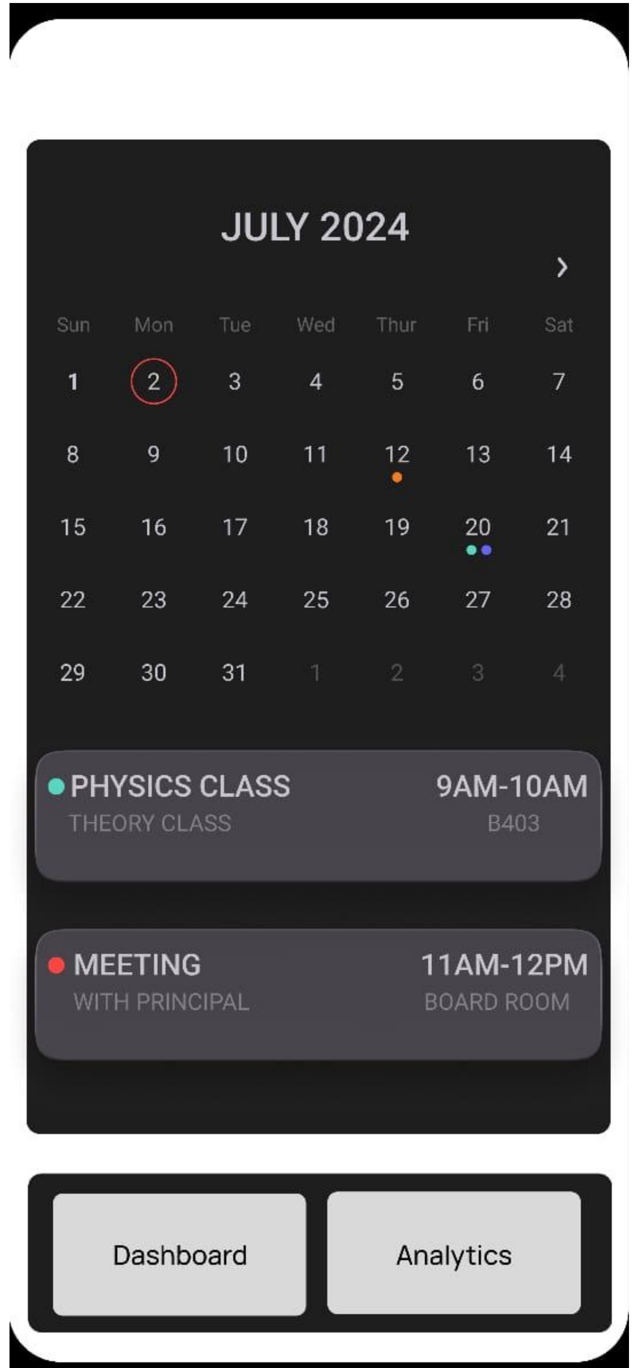

9:41
Smart - chair
Welcome Username. Now let's connect your Smartchair using:
Allow **Bluetooth** to find, connect to, and determin the relative positive of nearby devices?
Cancel
Allow
9:41
Smart- chair
Welcome Username. Now let's connect your wheelchair using:
Scanning your devices ...


9:41
Smart - chair
Welcome Username. Now let's connect your Smartchair using:
Dicoverable Bluetooth devices
Smart-chair 001
Connect
gayu
Connect
esha
Connect
dinesh
Connect
Gururaj
Connect
Scanning your devices ...
9:41
Verification Serial Number
Please enter the serial number from your product Smart - chair
Enter Serial number:
Enter here.....
Verify
Your health and comfort connected seamlessly


9:41
Welcome to your Dashboard
Your Average Streak🔥: 4200
YOUR SCORE
2800/5000
Gayu's Smart Chair
76BPM
Heartrate
66KG
Load
45MIN
Sit-Streak
490NITS
Ambient light
80%
Sit-Streak
Stress
Light Score
Remainders
Analytics


9:41
Analytics
Your Week
4750
2800
Previous week
This week
Mo
Tu
We
Th
Fr
Sa
sun
Dashboard
LeaderBoard


9:41
Leaderboard
Gayathri
5146
Dinesh
3551
Esha
2771
Archana 1124
Harini 1165
Gautham 875
Diksha 1000
Ajay 2000
Guru 500
Dashboard
Analytics


JULY 2024
Sun Mon Tue Wed Thur Fri Sat
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31 1 2 3 4
PHYSICS CLASS 9AM-10AM
THEORY CLASS B403
MEETING 11AM-12PM
WITH PRINCIPAL BOARD ROOM
Dashboard
Analytics

FUTURE WORKS

- Advanced Industry-Level Sensor Integration
- Enhanced Posture Checking
- Advanced Employee Monitoring System
- Integration with Wearable Technology
- AI and Machine Learning Integration
- Comprehensive Health and Wellness Programs
- Corporate Health Initiatives
- Customization and Personalization
- Remote Health Monitoring

References

1.IOT Based Smart Chair to Help with Back Pain

Shilpa shesham

30 Jun 2023-International Journal For Science Technology And Engineering-Vol. 11, Iss: 6, pp 665-669

2.Li, Y., and Aissaoui, R., "Smart Sensor, Smart Chair, Can it Predicts Your Sitting Posture?," IEEE International Symposium on Industrial Electronics, Vol. 4, pp. 2754-2759, July. 2006.

3.C. S. Oh, M. S. Seo, J. H. Lee, S. H. Kim, Y. D. Kim, and H. J. Park,"Indoor Air Quality Monitoring Systems in the IoT Environment," The Journal of Korean Institute of Communications and Information Sciences, vol. 40, no. 5, pp. 886-891, May, 2015.